

Experiment No. 1

Aim: Cryptography in Blockchain, Merkle root tree hash

Code:

```
from pymerkle import InmemoryTree
import math

# Create an empty tree
tree = InmemoryTree()

def getHeight(nLeaf):
    return math.ceil(math.log2(nLeaf)) if nLeaf>0 else 0

initial_hash=tree.get_state()

print("==== EMPTY TREE ====")
print("Root Hash :", tree.get_state().hex())
print("Size    :", tree.get_size())
print("Leaves   :", len(tree.leaves))
print("Height   :", getHeight(len(tree.leaves)) )

# Add some records
tree.append_entry(b"Apple")
tree.append_entry(b"Banana")
tree.append_entry(b"Orange")
tree.append_entry(b"Grapes")

print("\n==== AFTER ADDING RECORDS ====")
```

```
print("Root Hash :", tree.get_state().hex())
print("Size      :", tree.get_size())
print("Leaves    :", len(tree.leaves))
print("Height    :", getHeight(len(tree.leaves)) )
```

```
# using the hashlib
import hashlib
```

```
data1=b"Apple"
data2=b"Banana"
data3=b"Orange"
data4=b"Grapes"
```

```
print("Data:Apple,Banana, Orange, Grapes")
```

```
# Individual Hash
```

```
hash1=hashlib.sha256(data1).digest()
hash2=hashlib.sha256(data2).digest()
hash3=hashlib.sha256(data3).digest()
hash4=hashlib.sha256(data4).digest()
```

```
# Hash combine level 1
```

```
hash12=hashlib.sha256(hash1+hash2).digest()
hash34=hashlib.sha256(hash3+hash4).digest()
```

```
# Root hash
```

```
parent=hashlib.sha256(hash12+hash34).digest()
```

```
main=hashlib.sha256(parent+initial_hash).digest()
```

```
print(
```

```
f"
```

```
hash1({hash1.hex()})
```

```
|-----hash12({hash12.hex()})
```

```
hash2({hash2.hex()})
```

```
parent({main.hex()})
```

```
hash1({hash3.hex()})
```

```
|-----hash34({hash34.hex()})
```

```
hash1({hash4.hex()})
```

```
")
```

Output:

```
parent(205906d3f4c56515cc5382f8656a1ff5a51fa041bd69f6257cb841c4ac6dd2c8)
|-----hash12(8ee211056dbf00f02a4c6dd701908681d7a257a6c01d01126335c215e2f6f8a3)
|-----hash1(f223faa96f22916294922b171a2696d868fd1f9129302eb41a45b2a2ea2ebbfd)
|-----hash2(f9782dd7999dc14b39c1329735e6e4ef72e77a3cf5fa32f2f57bf8d5493f0fc5)
|-----hash1(78e7771b8b46e11ddb34ba48887e1330525215f96d94778980d1186e6f09f6b4)
|-----hash34(ced97a9289969869db8ea08418e8a4c4b6b8fcf44d1dd5d407c407ac33824fe7)
|-----hash1(d5fab6d3de1d6a40c11a2e92f11e78b1656cd09dee7a63f4623906694ff0e3ff)
```

```
● (.venv) PS C:\Users\Mukesh kushwaha\Documents\College\Codes\sem-7\BC Codes> python exp1.py
===== EMPTY TREE =====
Root Hash : e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
Size      : 0
Leaves    : 0
Height    : 0

===== AFTER ADDING RECORDS =====
Root Hash : 9b2613c37cca2a201ebb02d1cb33a7cbc9c2db534e4edf9dd83e28bd8683e7f2
Size      : 4
Leaves    : 4
Height    : 2
Data:Apple,Banana, Orange, Grapes
```